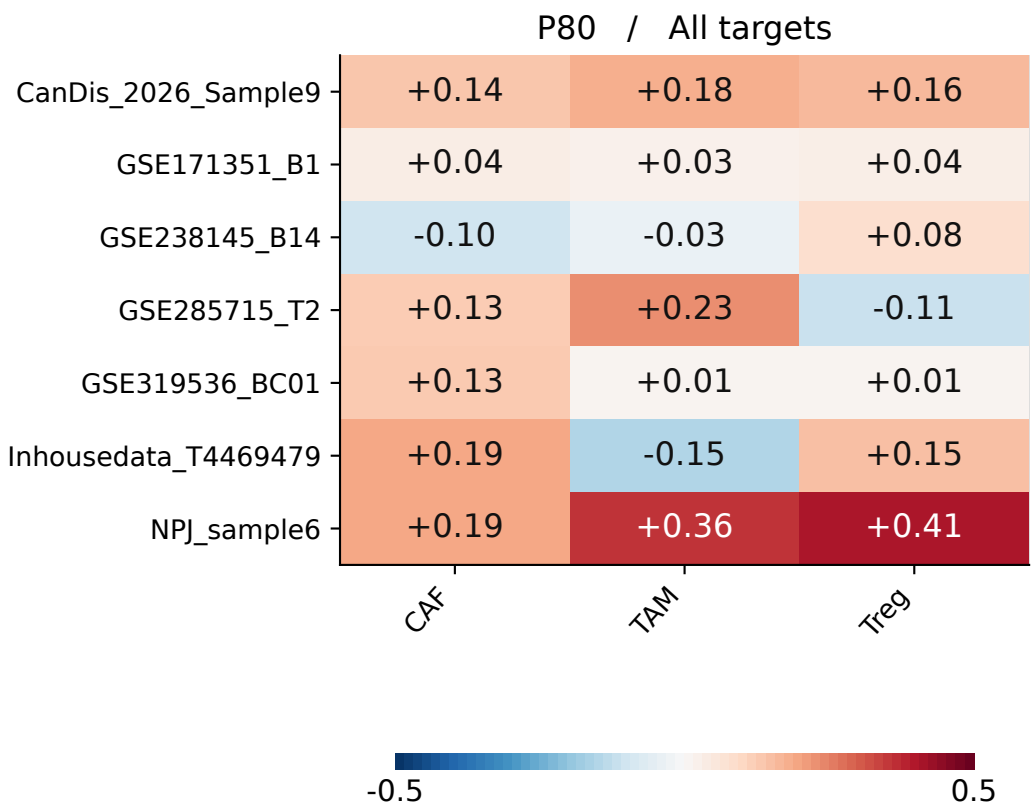
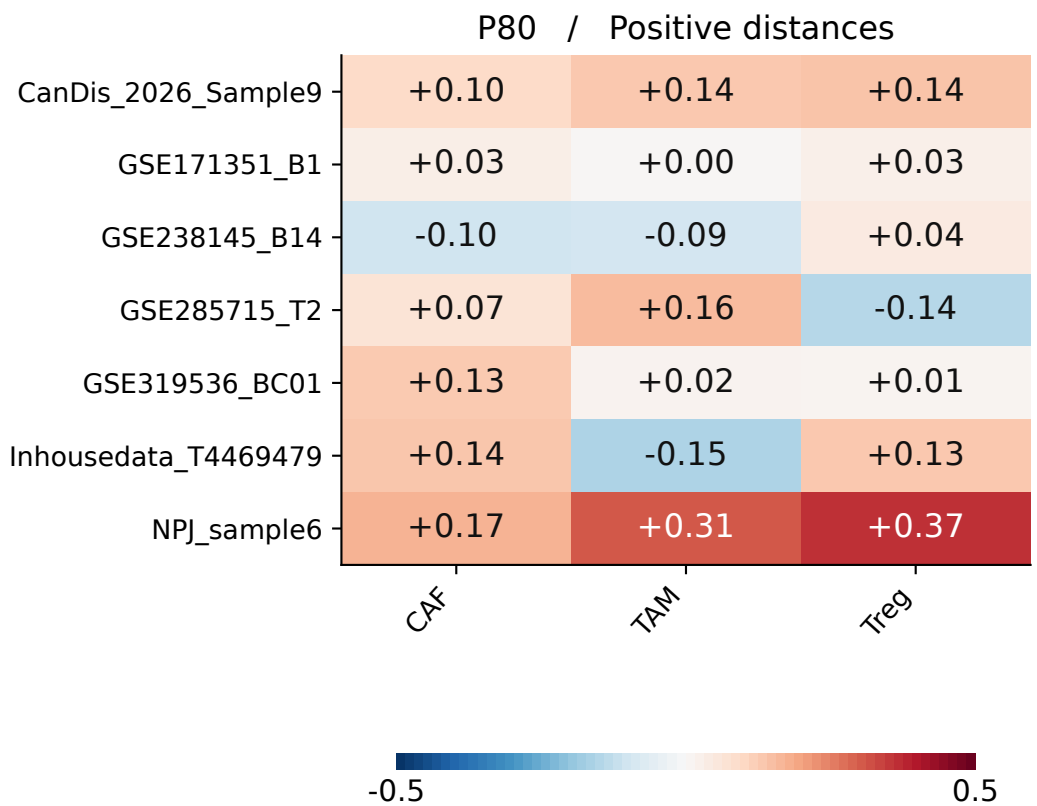
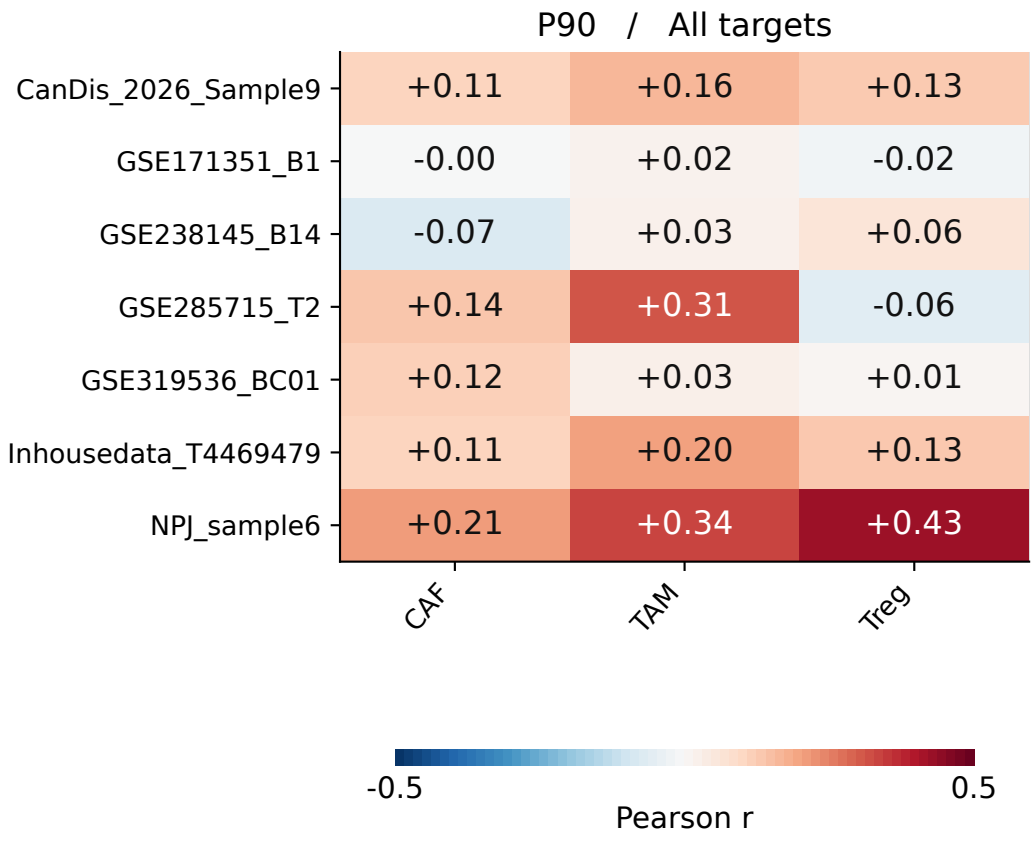
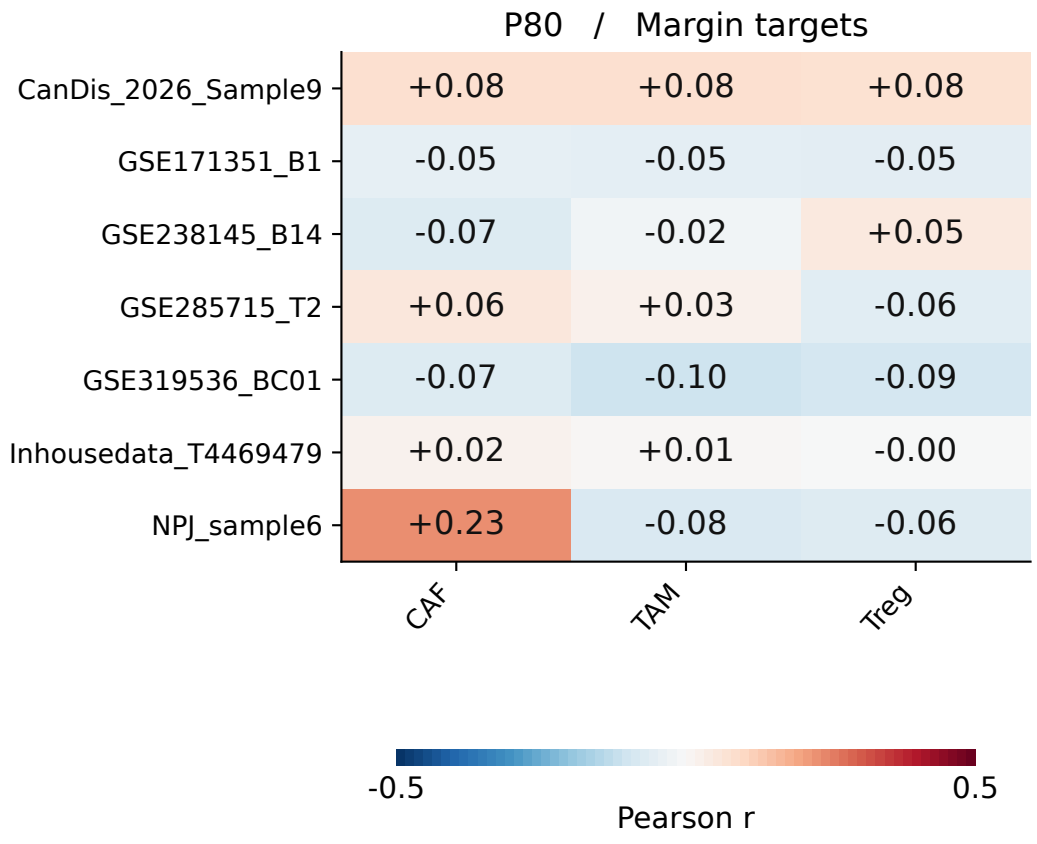
MP10 distance pilot | Seven sections

Section-level Pearson correlations. Positive r means higher mixed-spot MP10 farther from target.



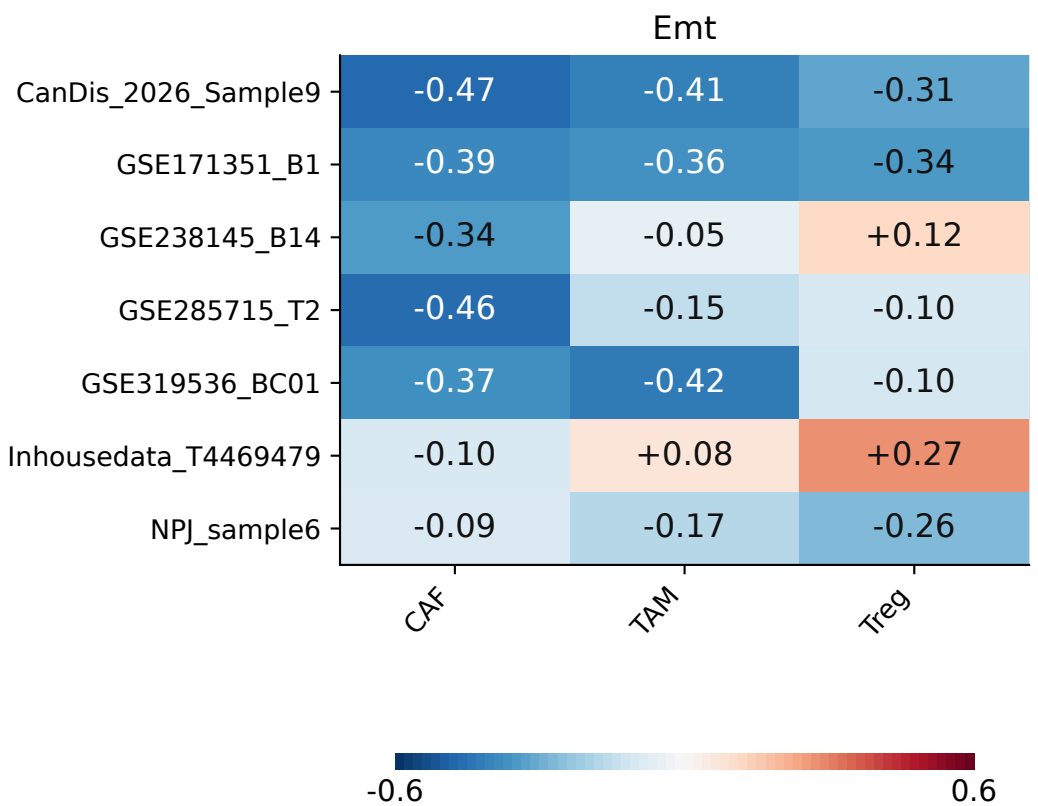


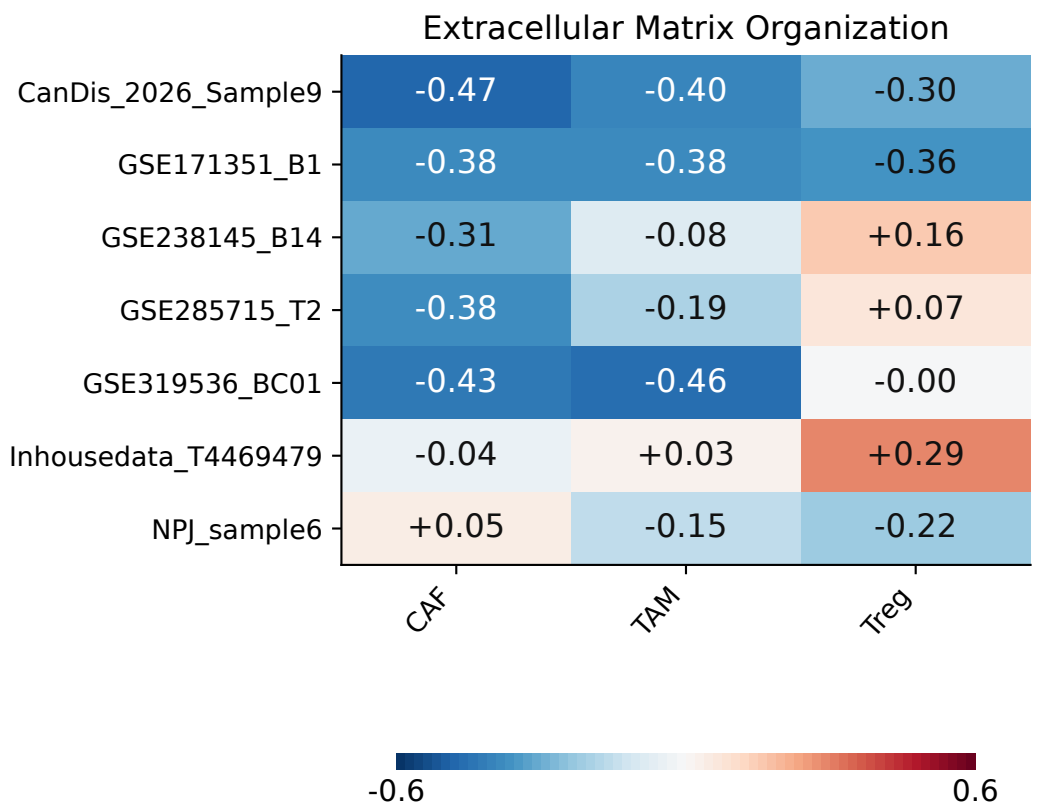


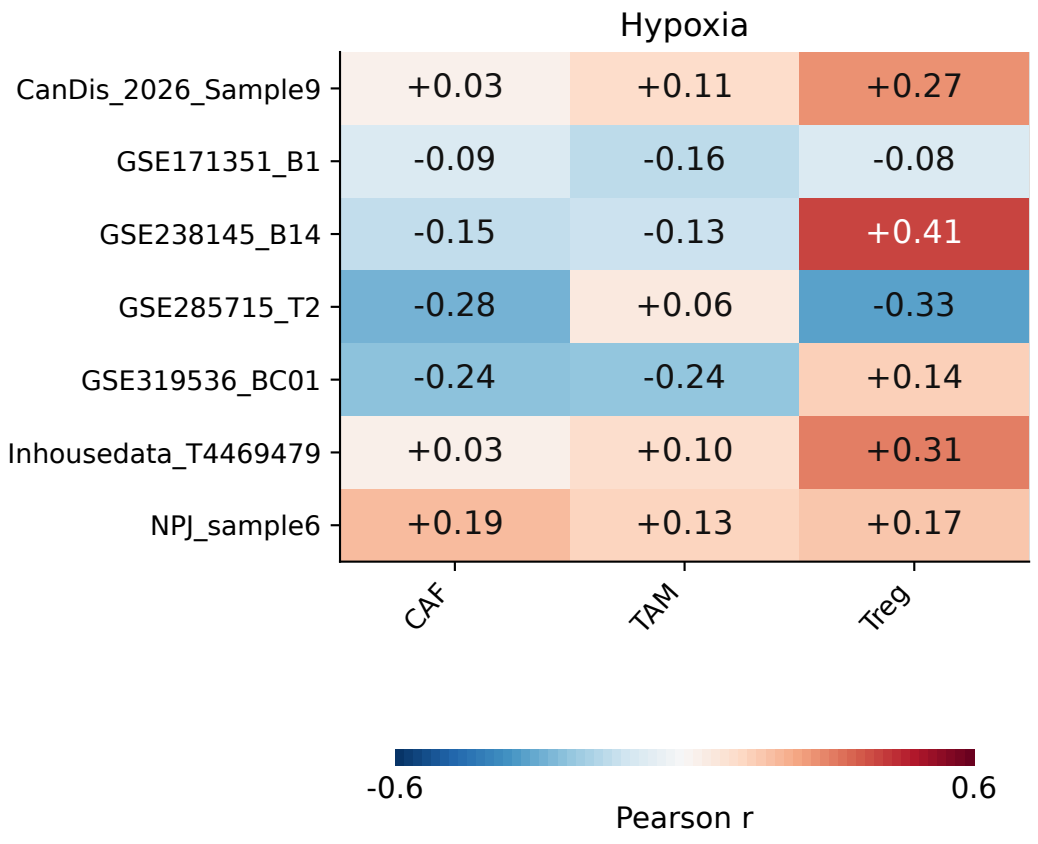


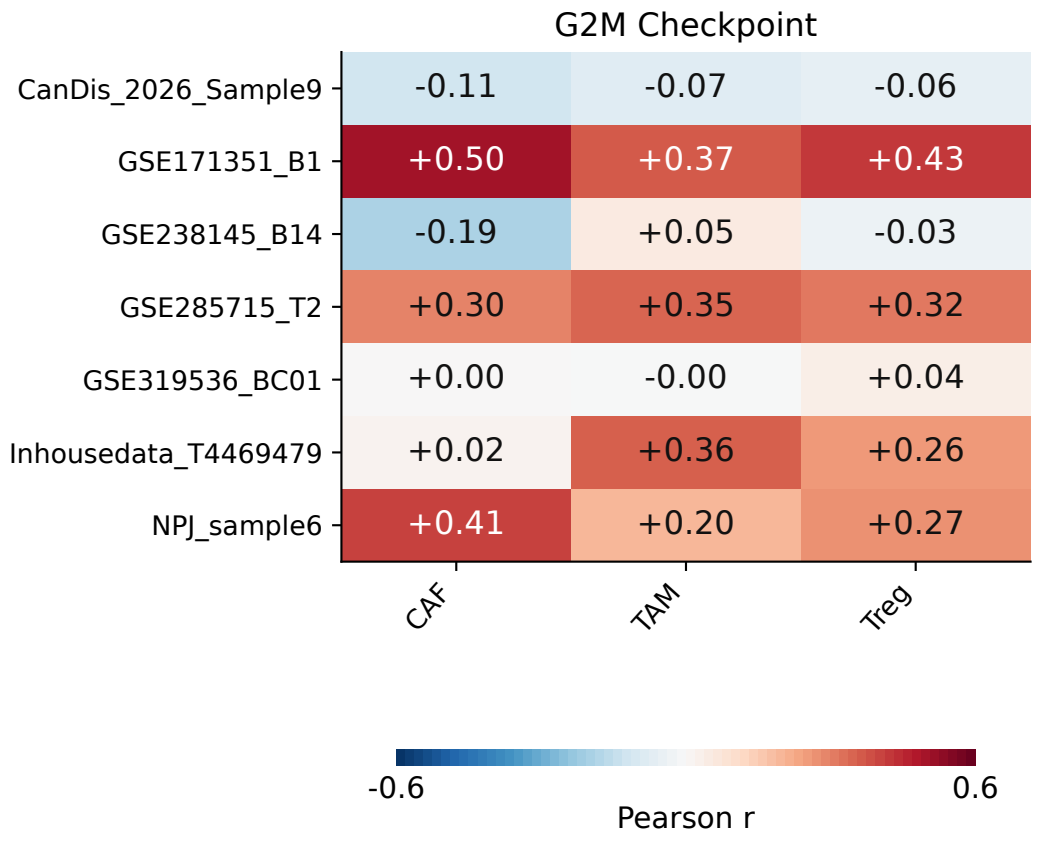
Spot transcriptional programs | Distance gradients

Pearson r within Epi queries • full-tissue P80 targets • color cap ±0.6



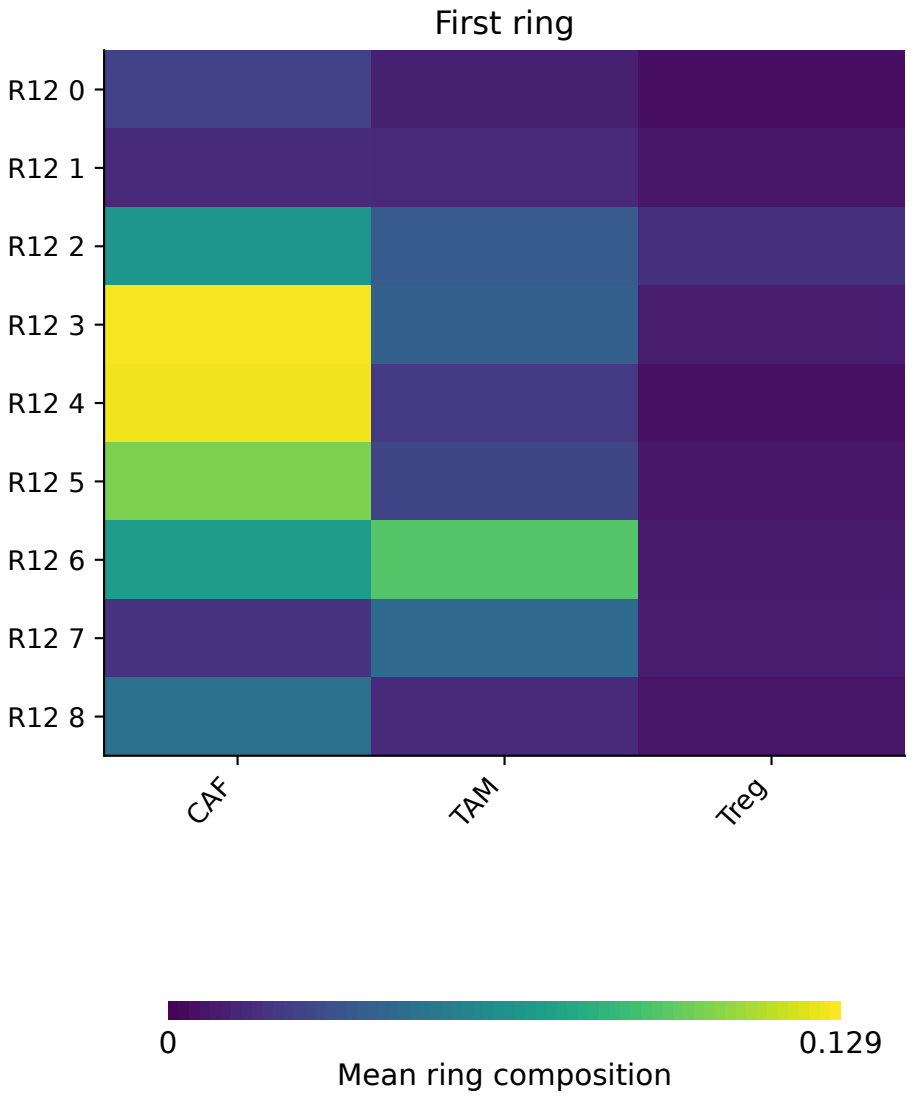


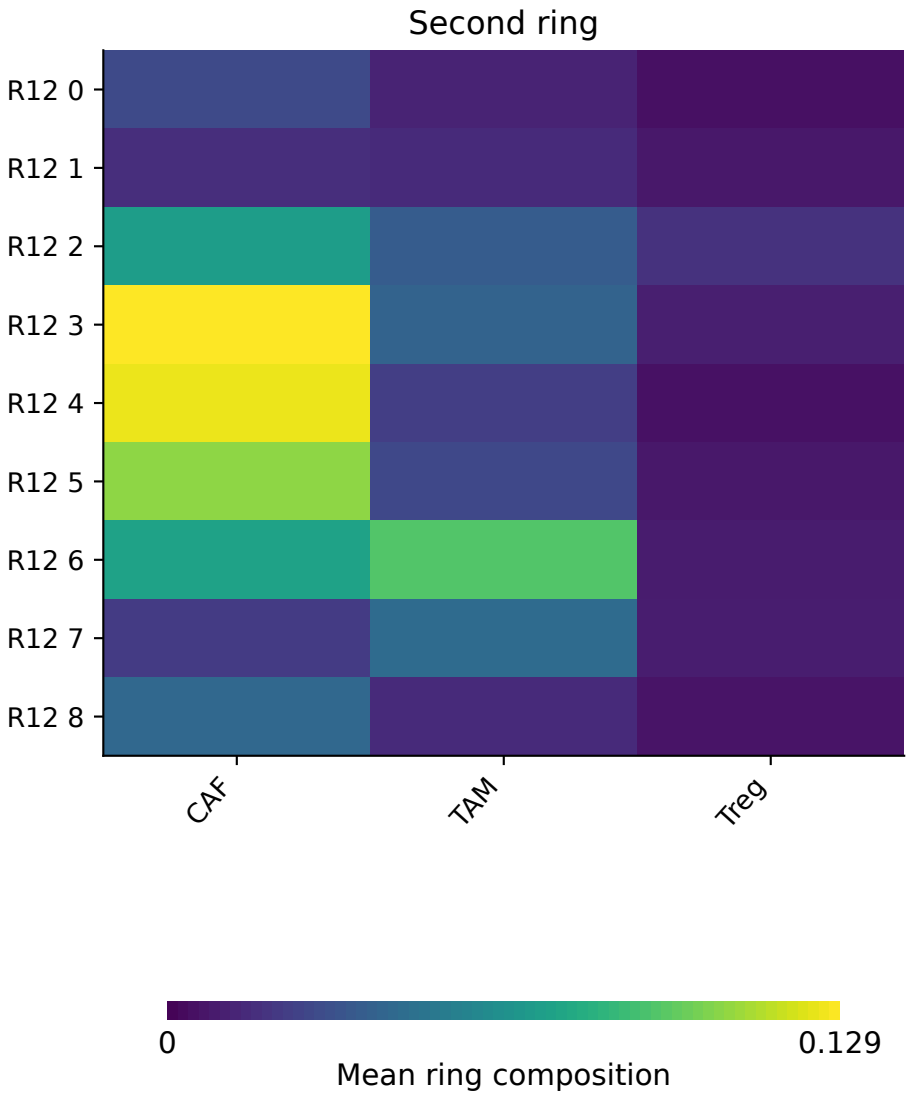




Accepted R12 environments | Reference ring profiles

Original SAMPLE_R12_LEIDEN neighborhood profile • mean relative q05 composition in R1 and R2

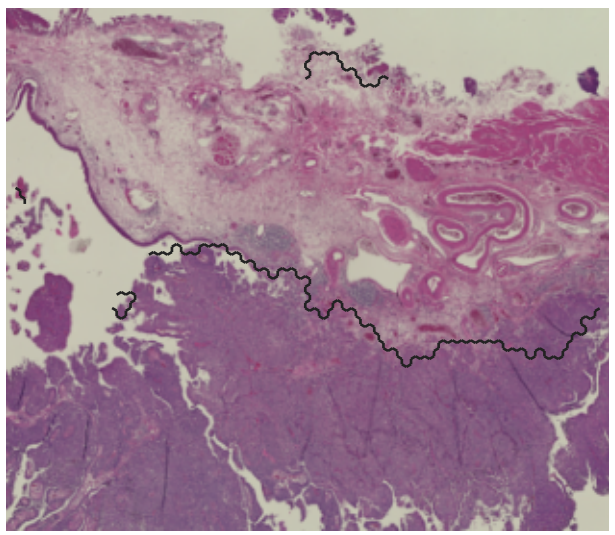




NPJ_sample6 | Tissue context

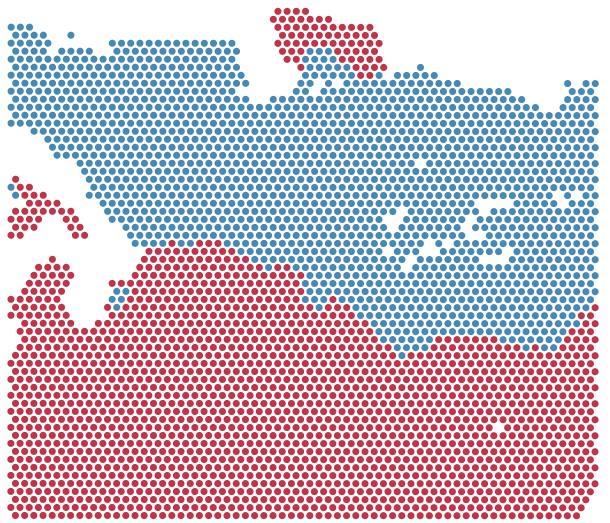
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



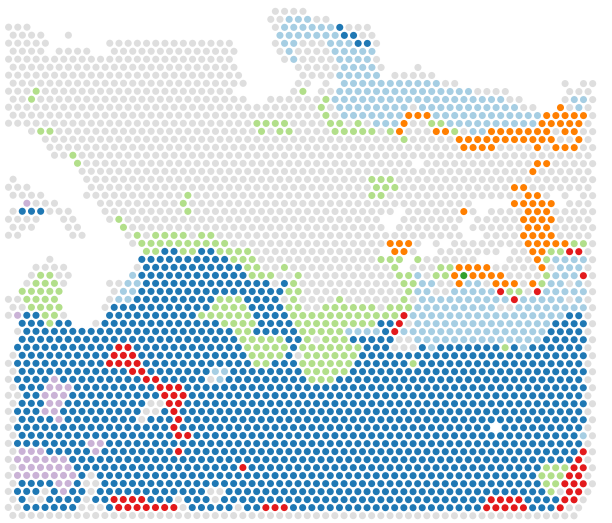
Black line: observed candidate outer edge

Final candidate location



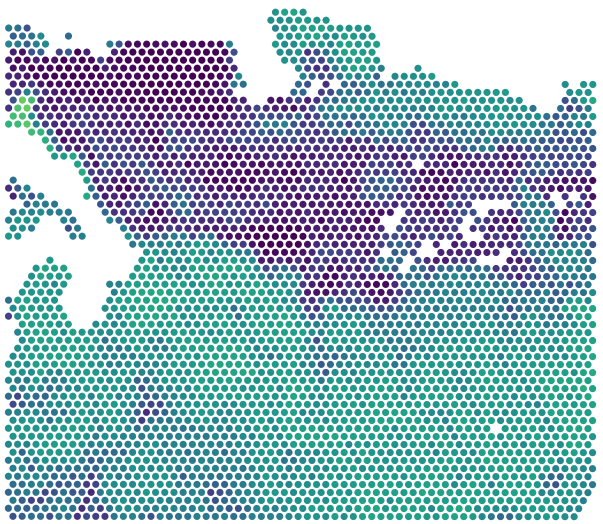
inside outside

Accepted R12 (original 0–8)



0 2 5 7 8
1 3

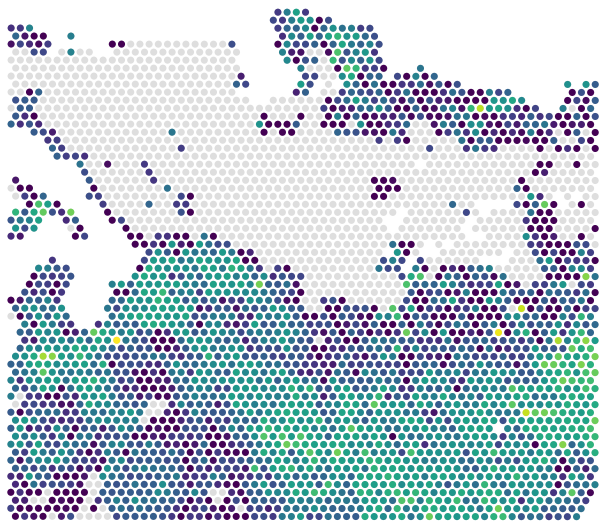
Epithelial support



0 1

Fraction of all 30

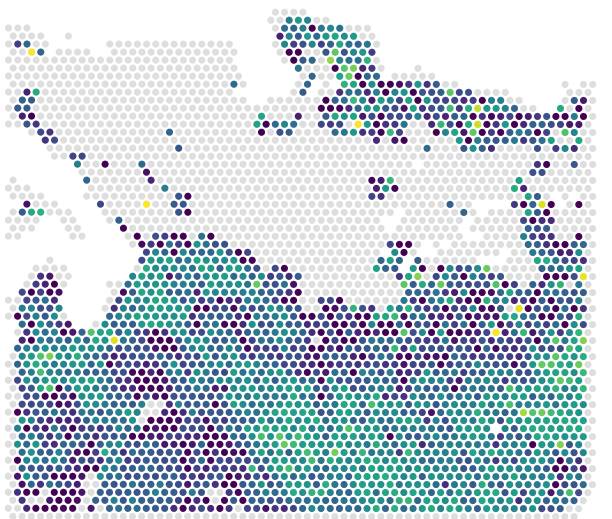
MP10 in Epi queries



0 0.149

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

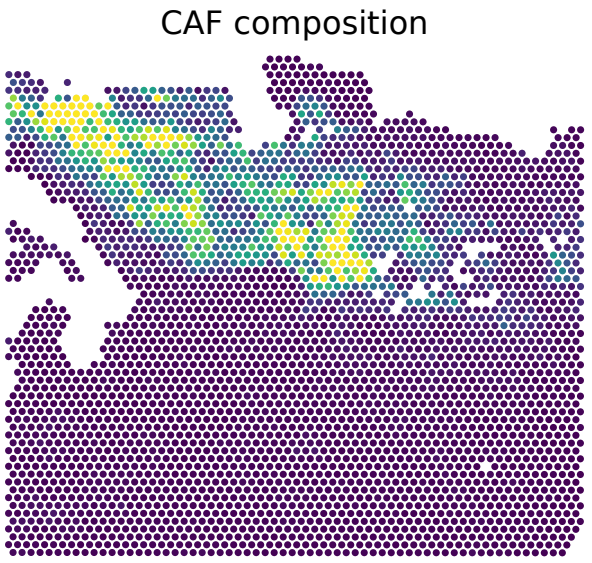


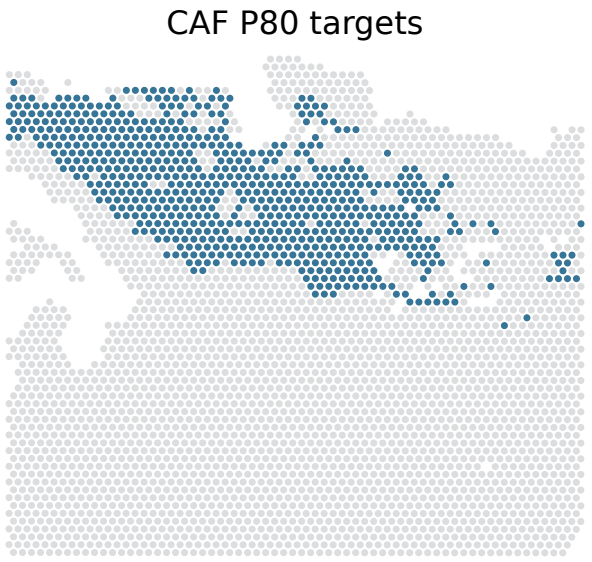
0 0.145

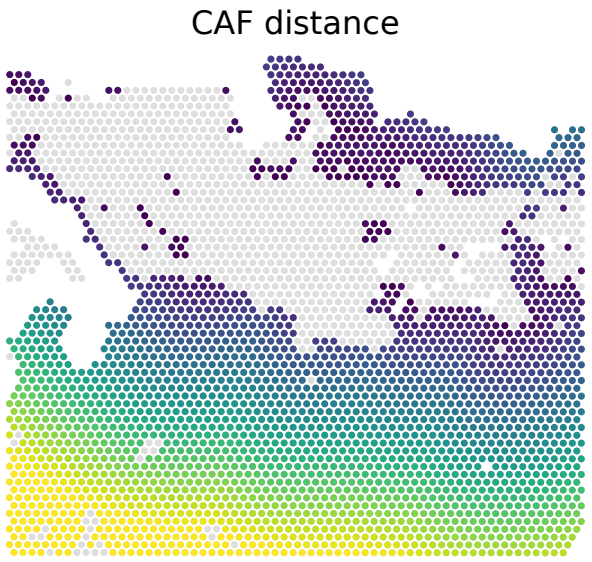
Matched-model AUCell

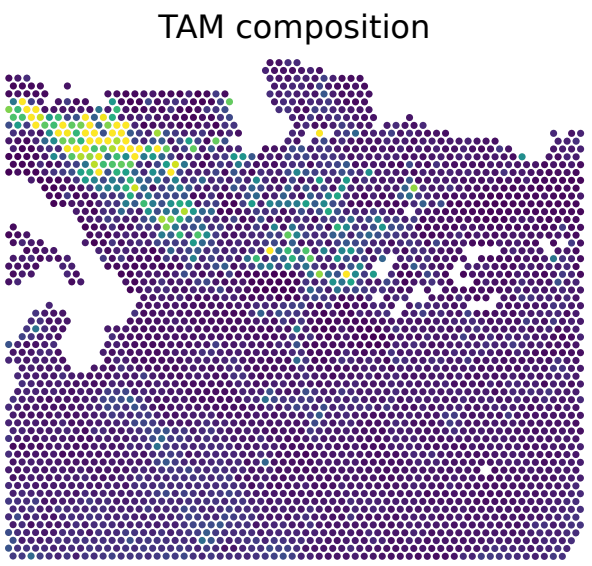
NPJ_sample6 | Candidate neighborhoods

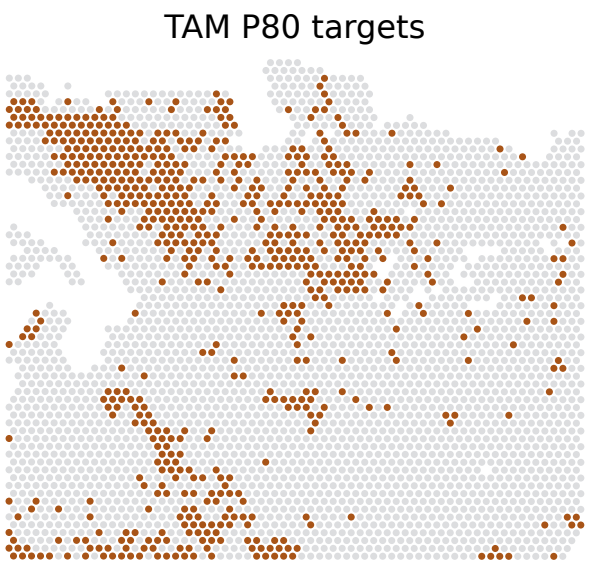
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

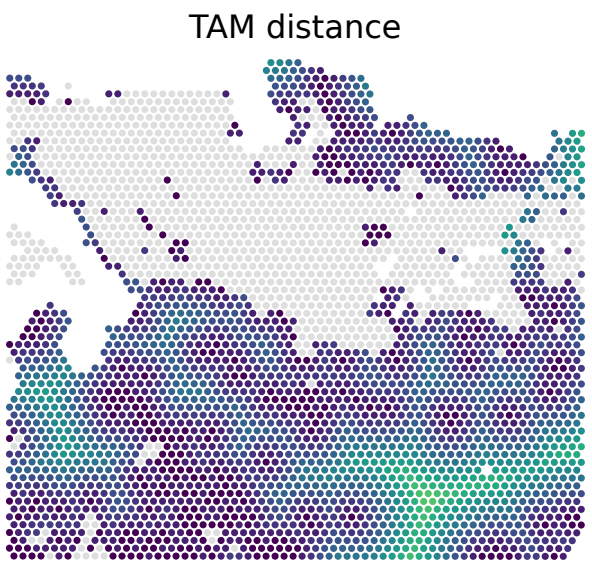


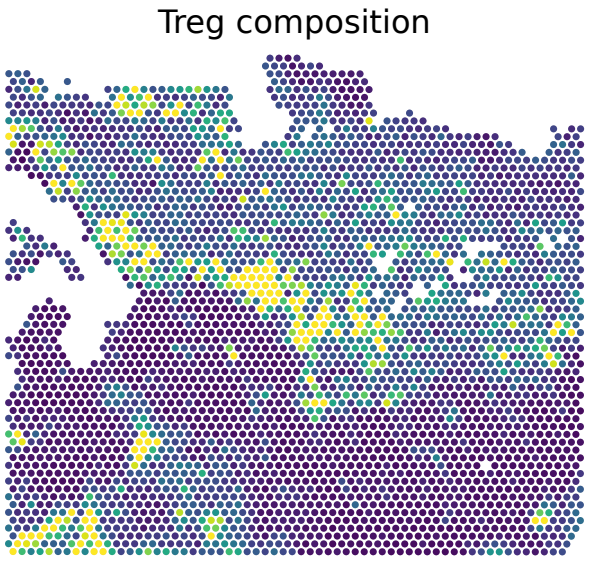


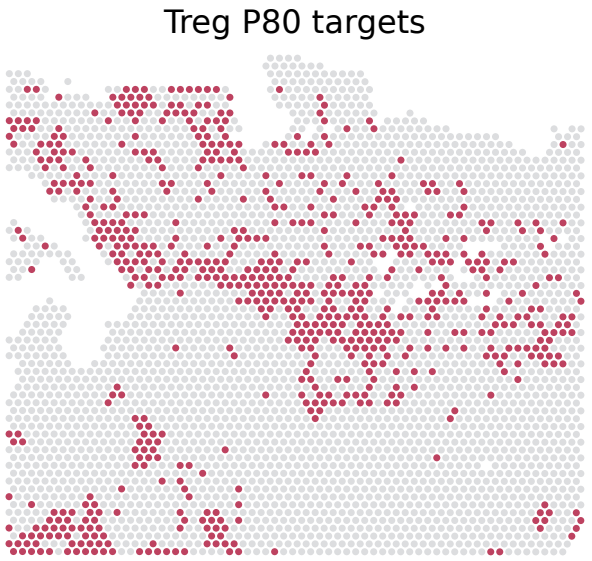


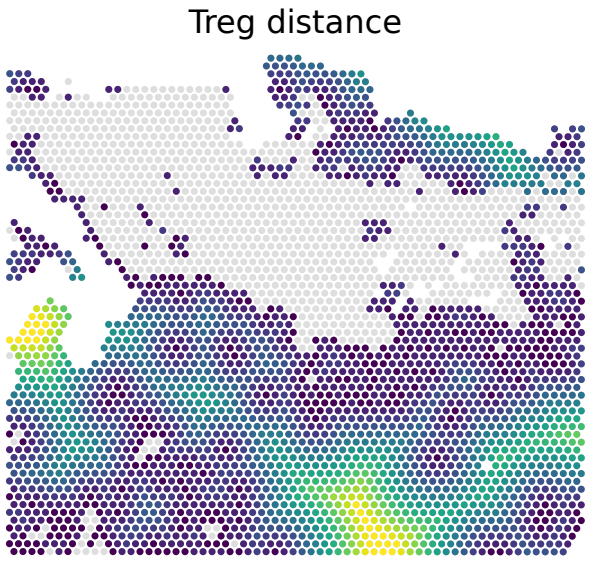






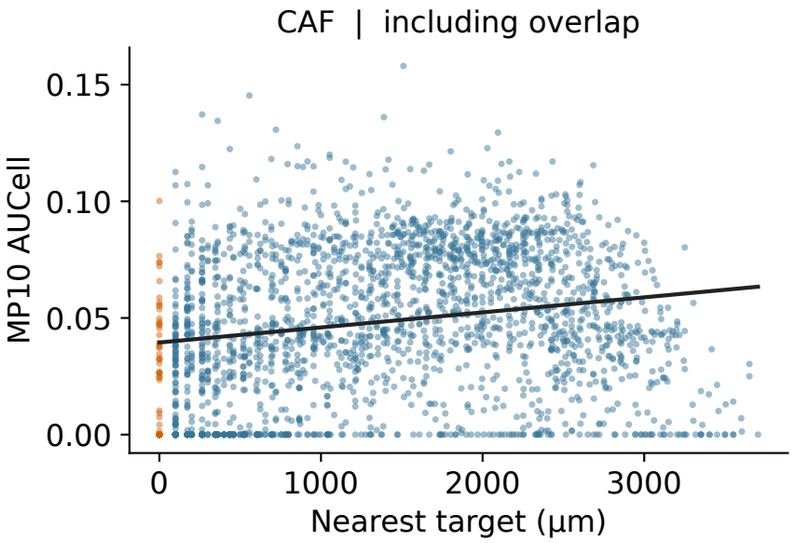




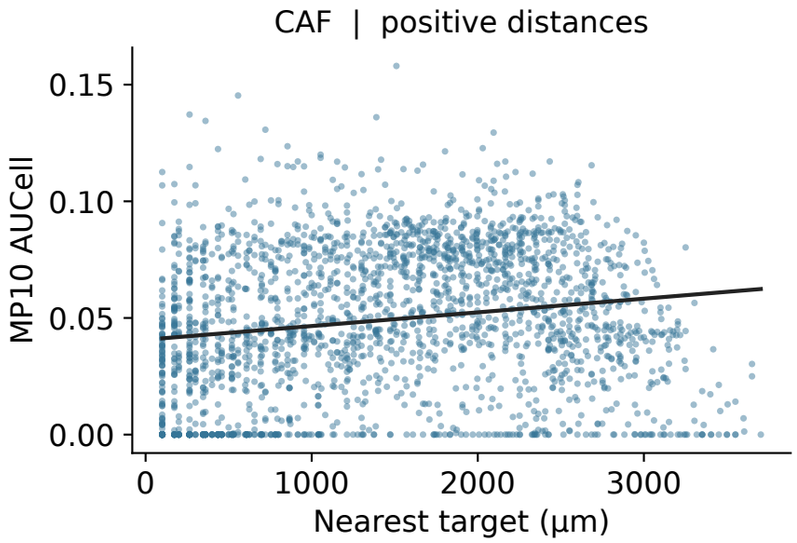


NPJ_sample6 | MP10 versus nearest target

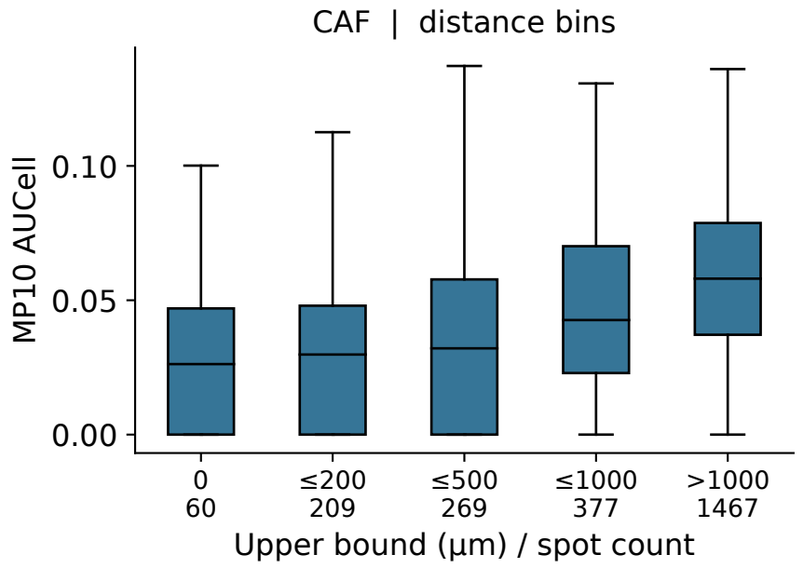
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



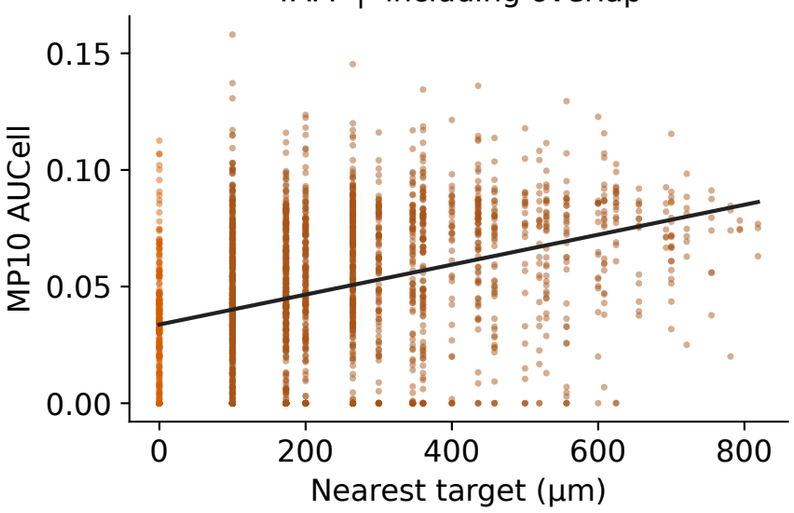
r +0.19 ρ +0.22
n 2382/2414 queries • 700 targets



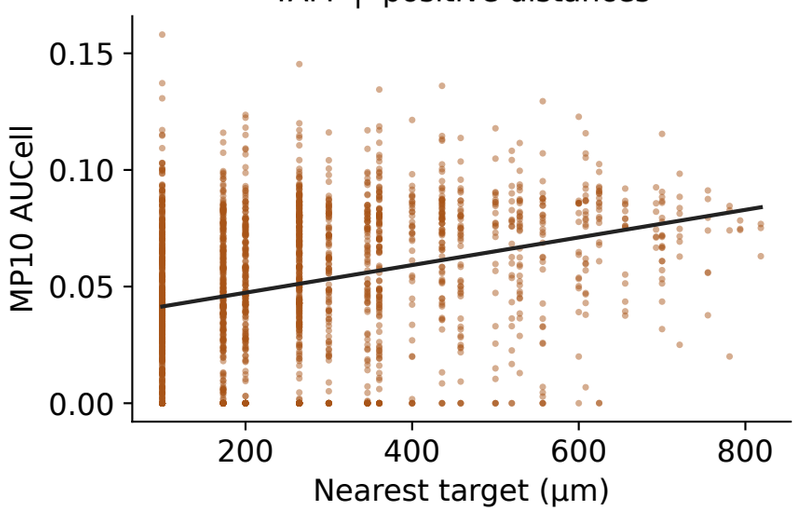
r +0.17 ρ +0.20
n 2322/2414 queries • 700 targets



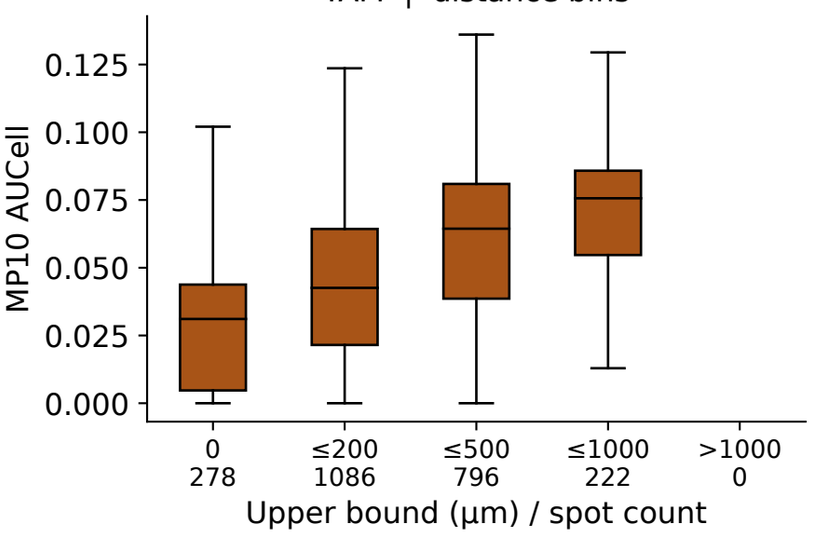
Orange = zero distance
Line = within-section linear fit



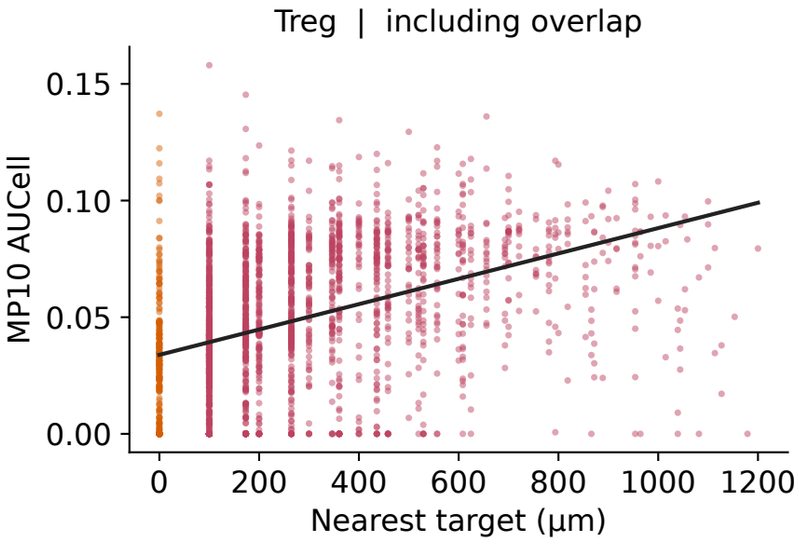
r +0.36 ρ +0.39
n 2382/2414 queries • 700 targets



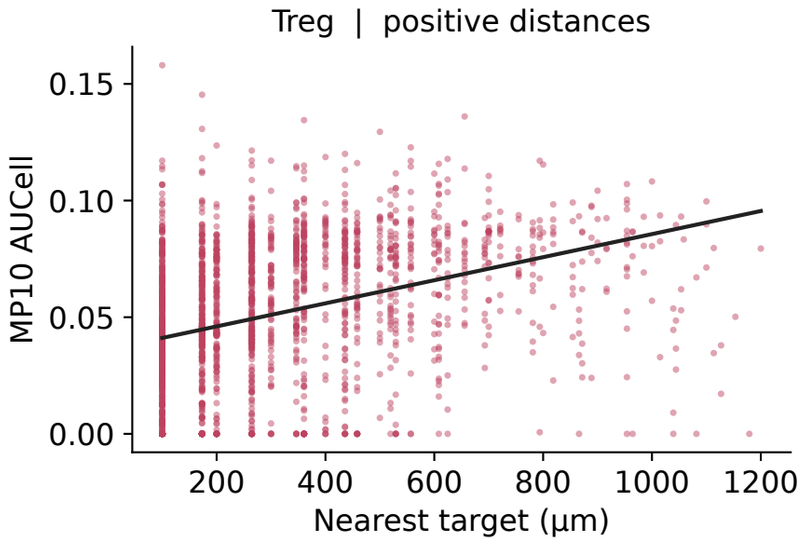
r +0.31 ρ +0.34
n 2104/2414 queries • 700 targets



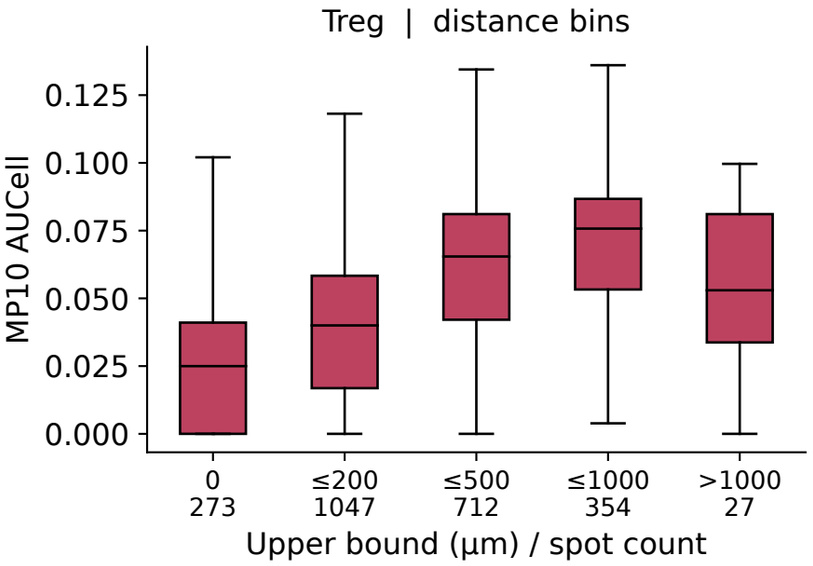
Orange = zero distance
Line = within-section linear fit



r +0.41 ρ +0.48
n 2413/2414 queries • 700 targets



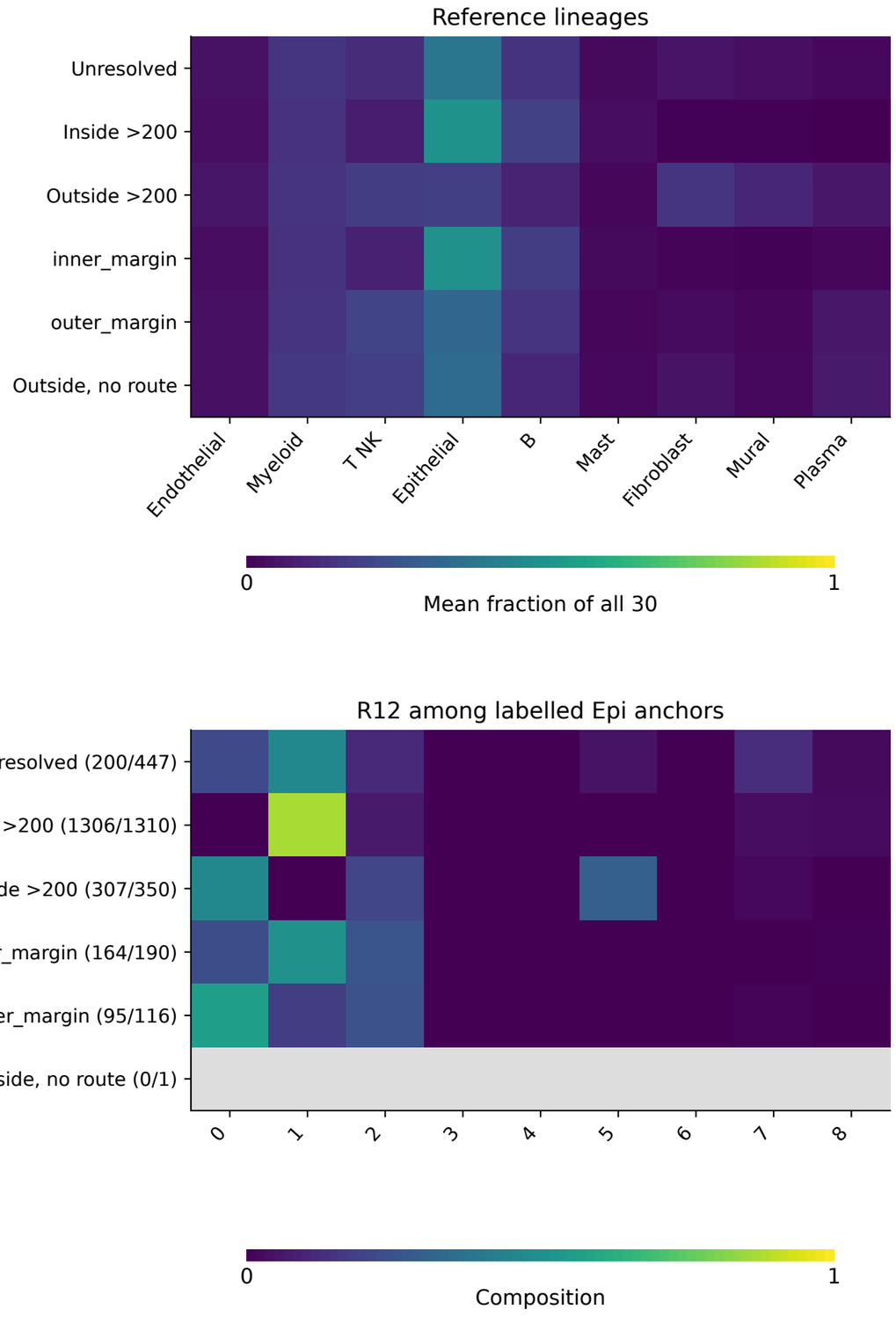
r +0.37 ρ +0.44
n 2140/2414 queries • 700 targets

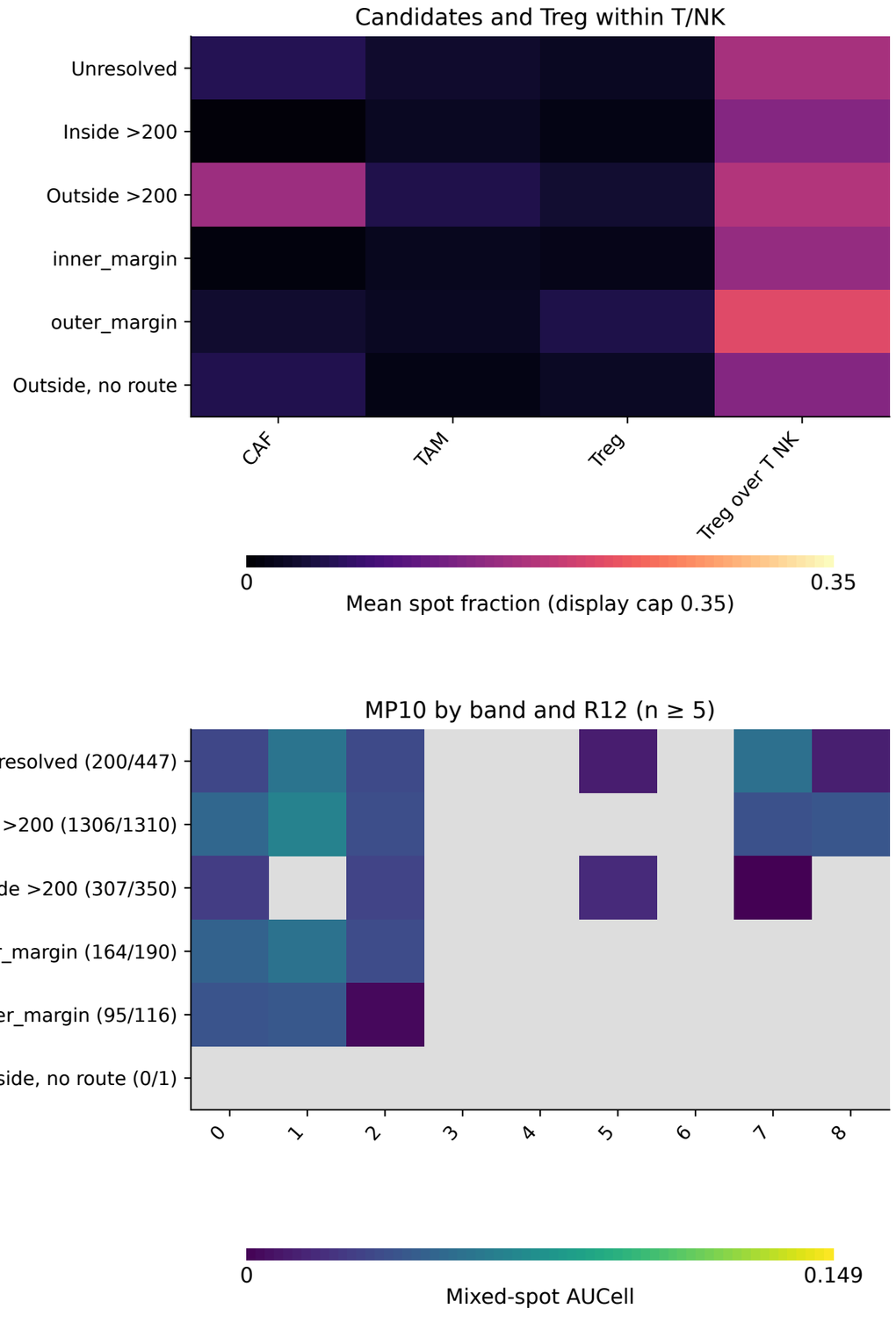


Orange = zero distance
Line = within-section linear fit

NPJ_sample6 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable

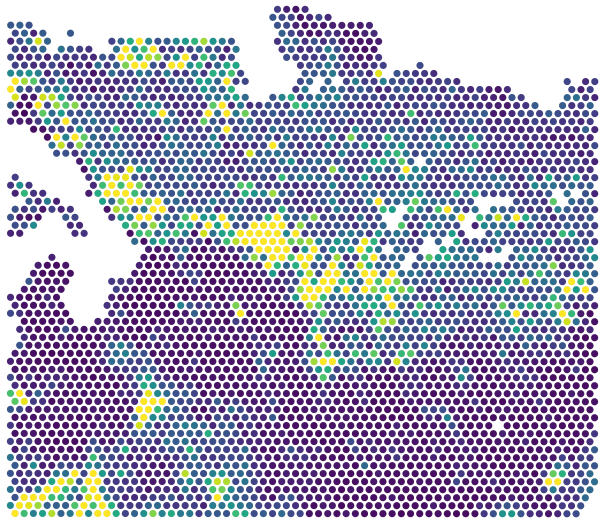




NPJ_sample6 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across pilot sections

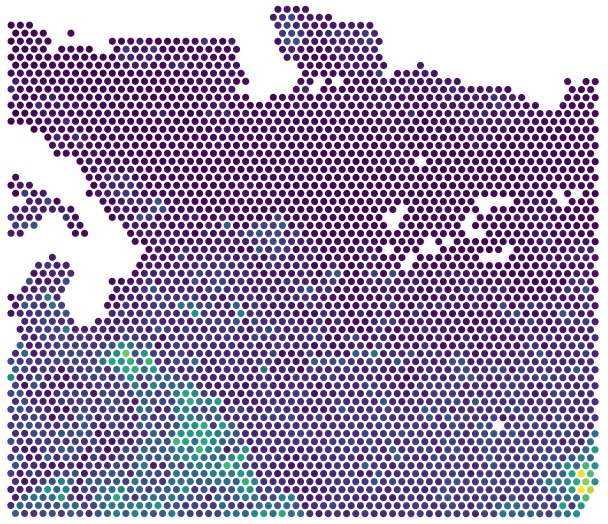
Treg composition



0 0.0901

Fraction of all 30

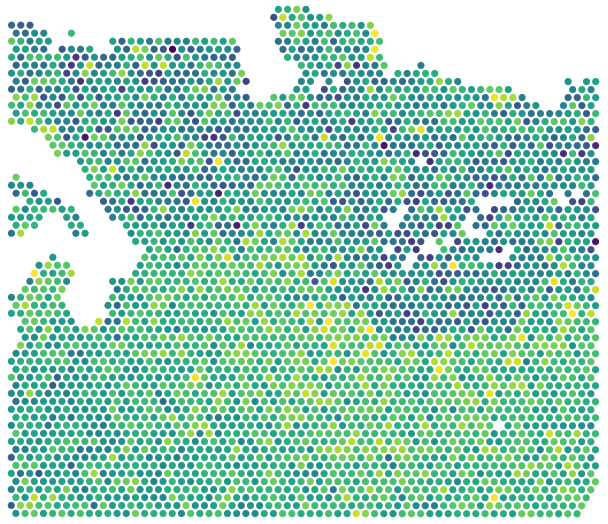
Treg estimated abundance



0 2.71

Posterior mean estimate

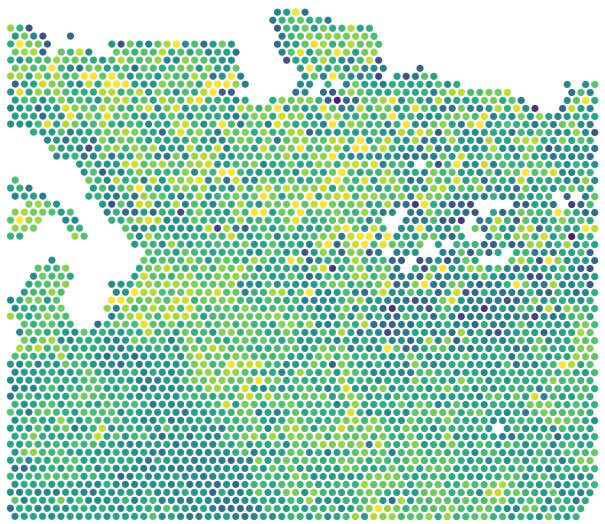
Treg Resting Vs Activated Dn



0 0.0795

UCell score

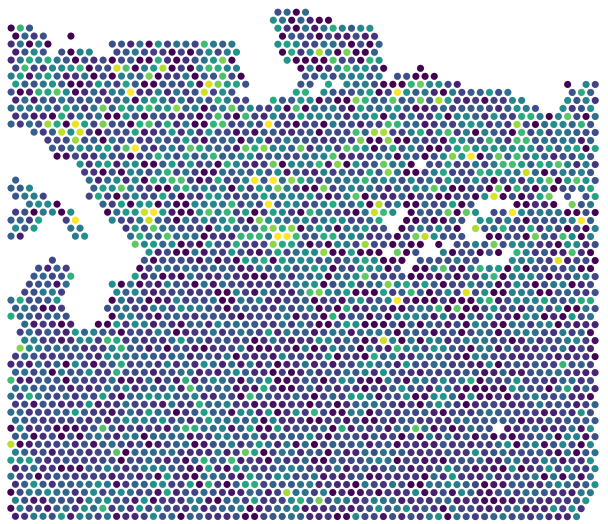
Treg Resting Vs Activated Up



0 0.063

UCell score

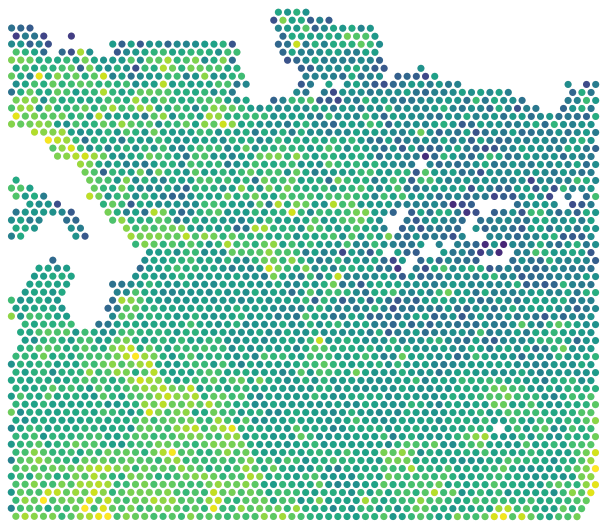
Co Inhibition By Ctla4



0 0.15

UCell score

Il2 Stat5 Signaling

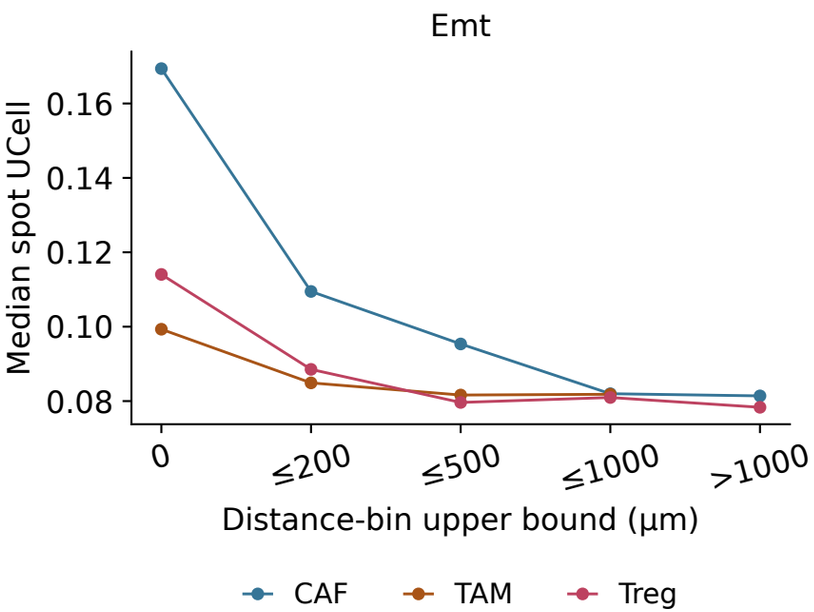


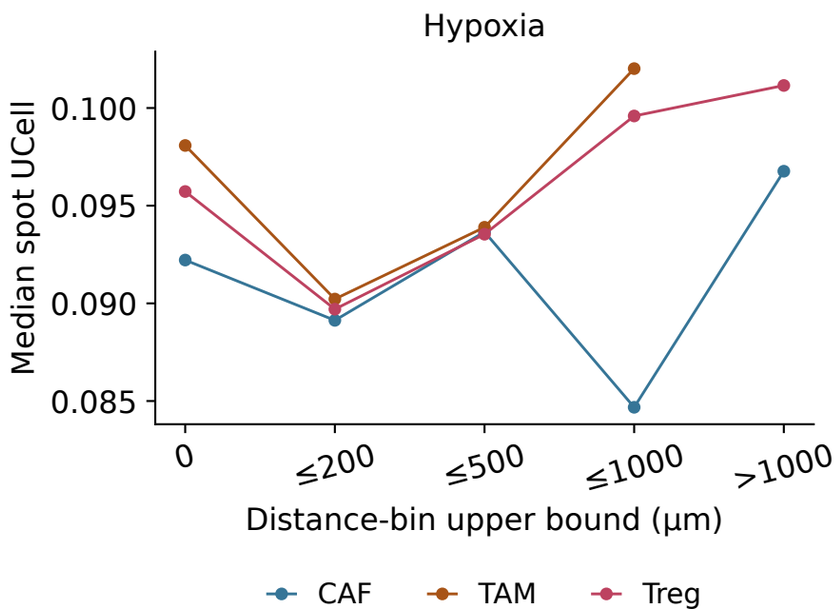
0 0.144

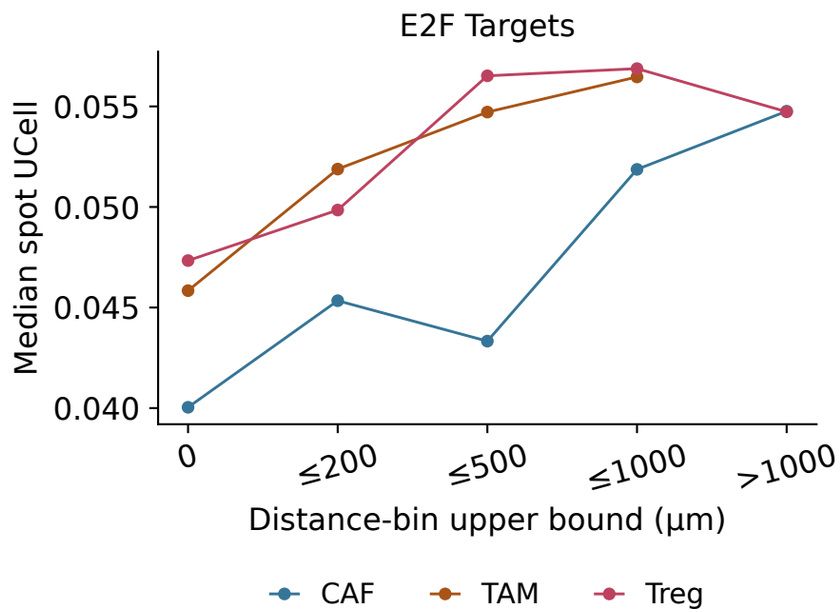
UCell score

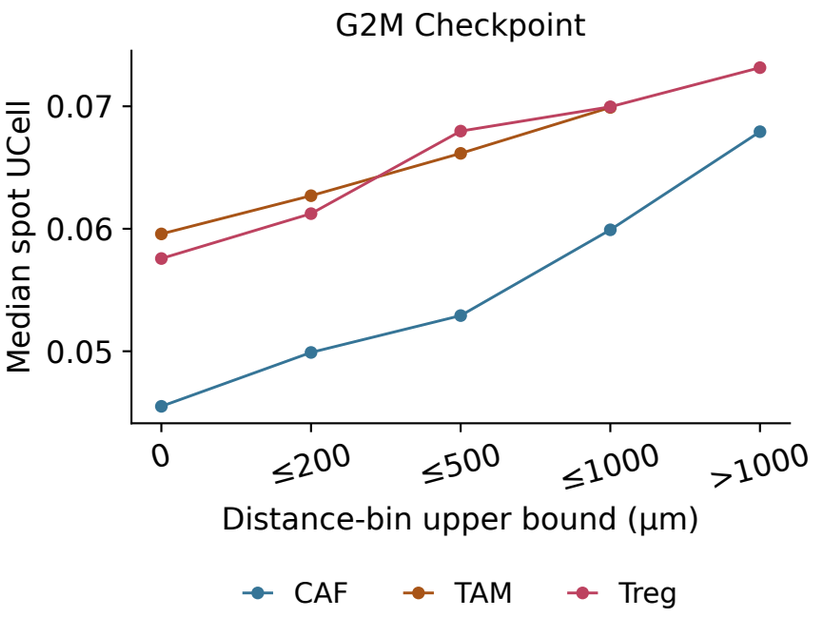
NPJ_sample6 | Programs by target distance

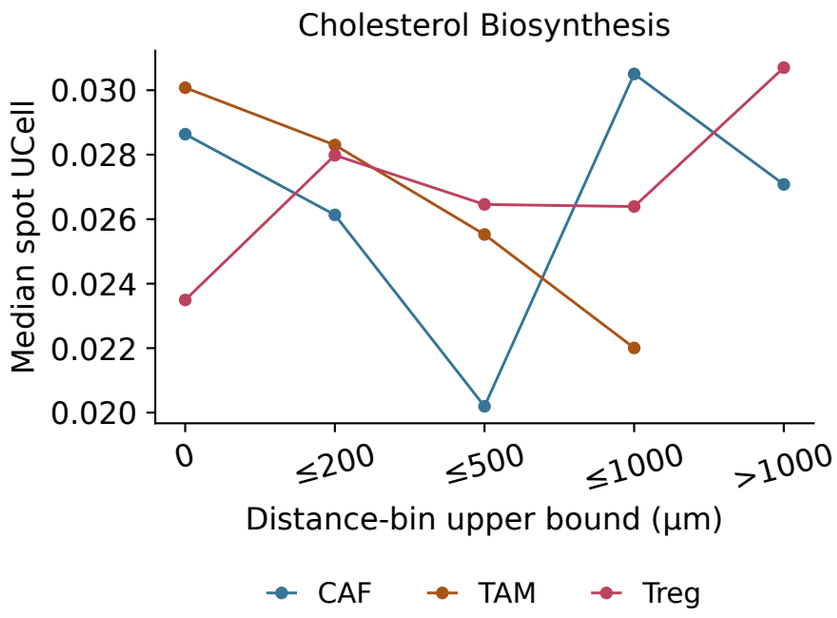
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

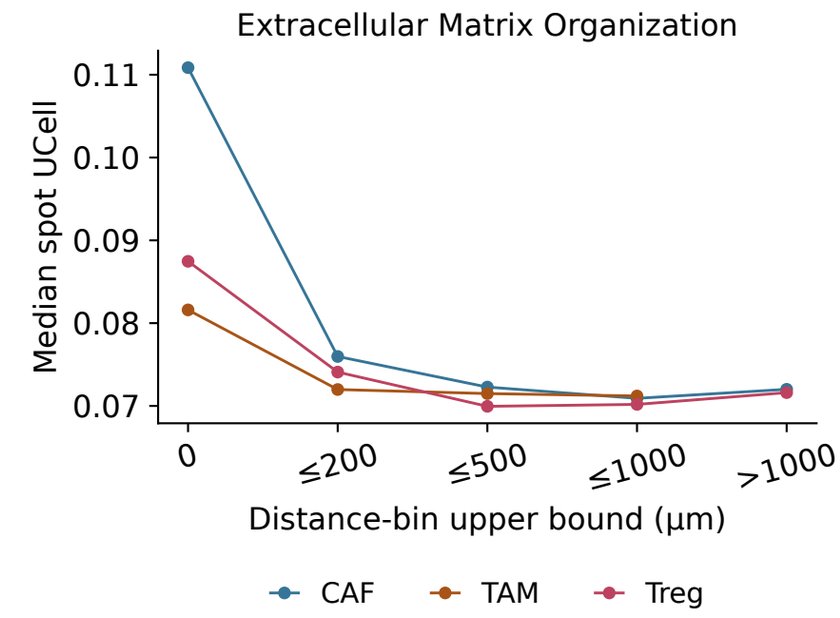






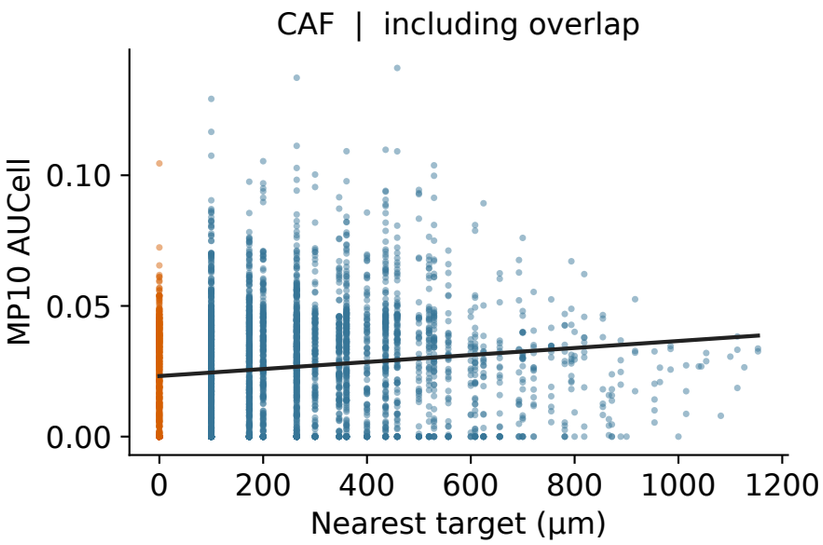




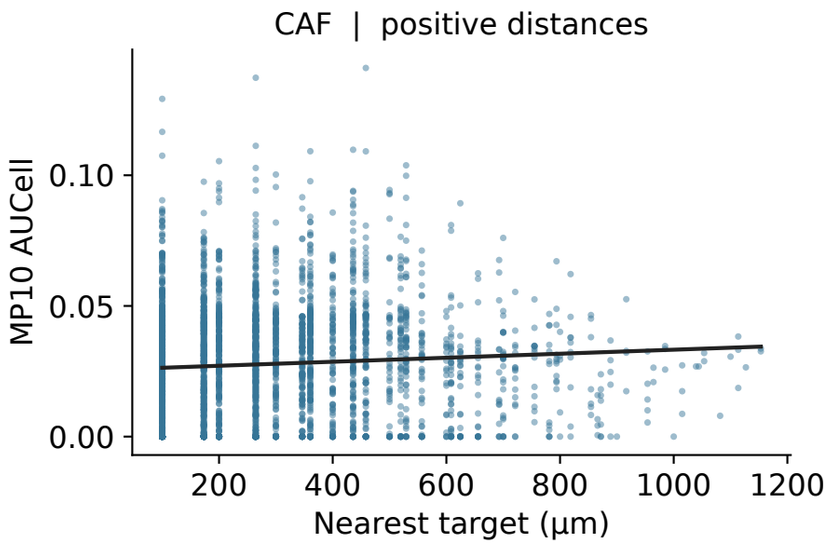


GSE285715_T2 | MP10 versus nearest target

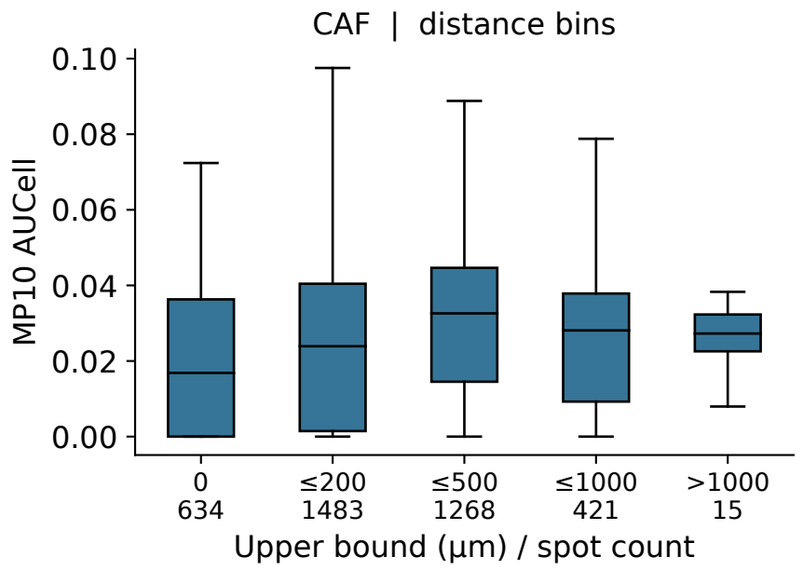
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



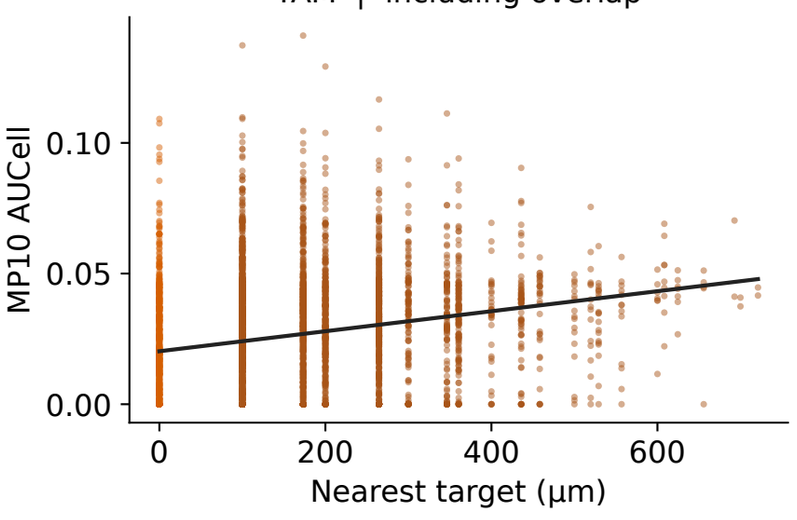
r +0.13 ρ +0.18
n 3821/3821 queries • 798 targets



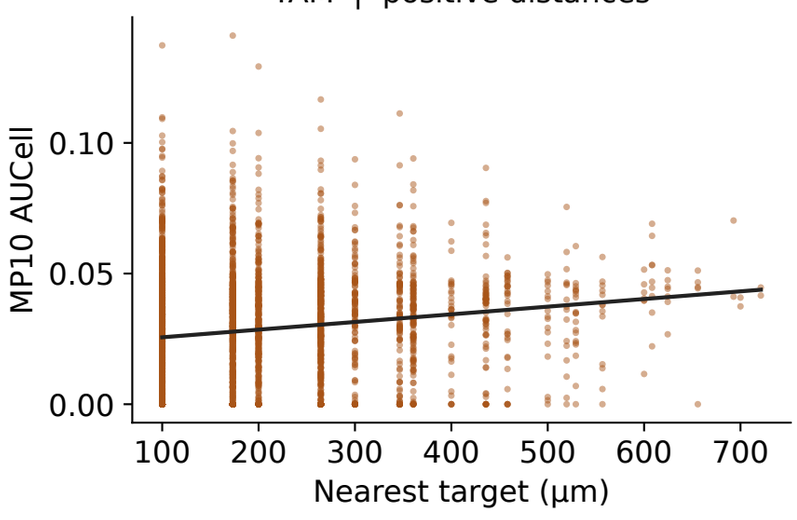
r +0.07 ρ +0.13
n 3187/3821 queries • 798 targets



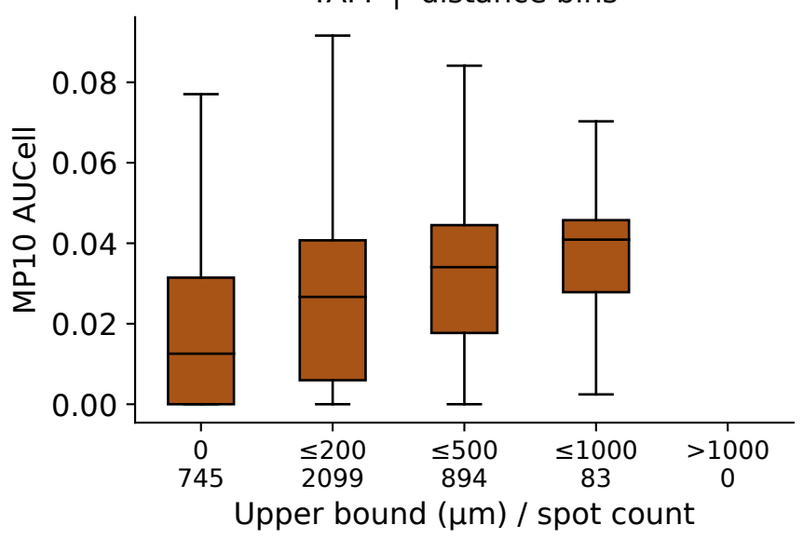
Orange = zero distance
Line = within-section linear fit



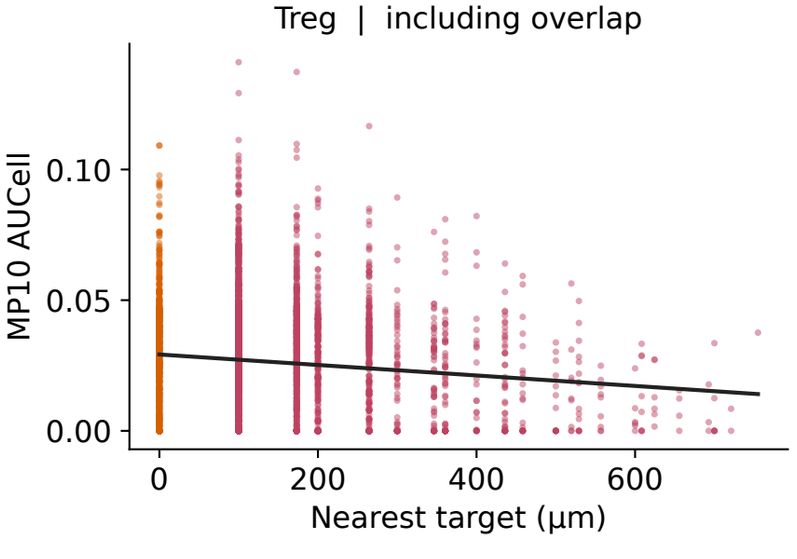
r +0.23 ρ +0.27
n 3821/3821 queries • 798 targets



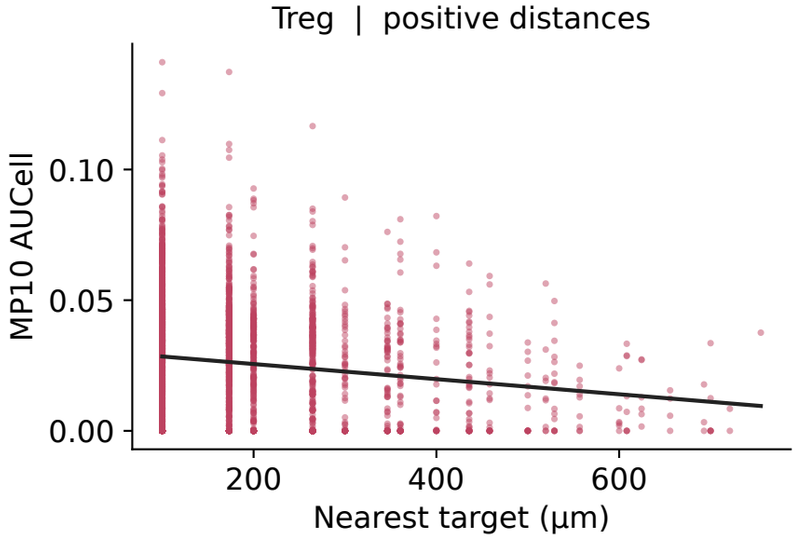
r +0.16 ρ +0.19
n 3076/3821 queries • 798 targets



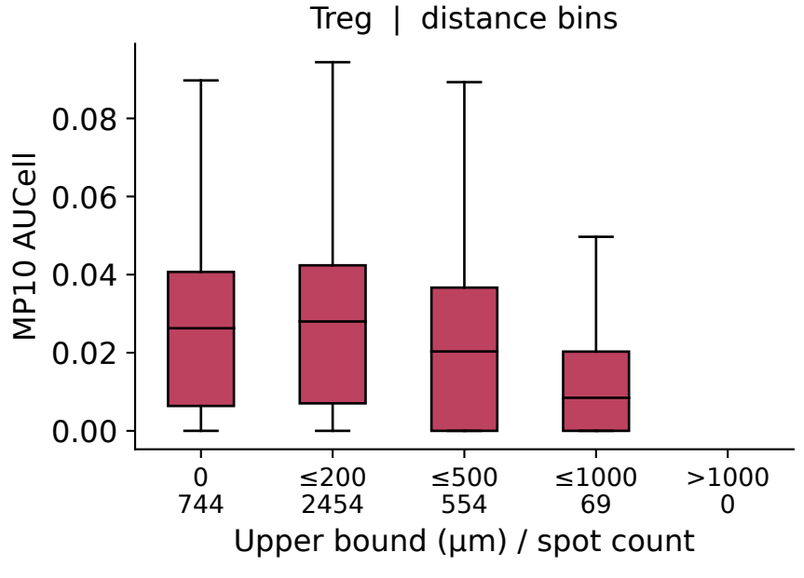
Orange = zero distance
Line = within-section linear fit



r -0.11 ρ -0.08
n 3821/3821 queries • 798 targets



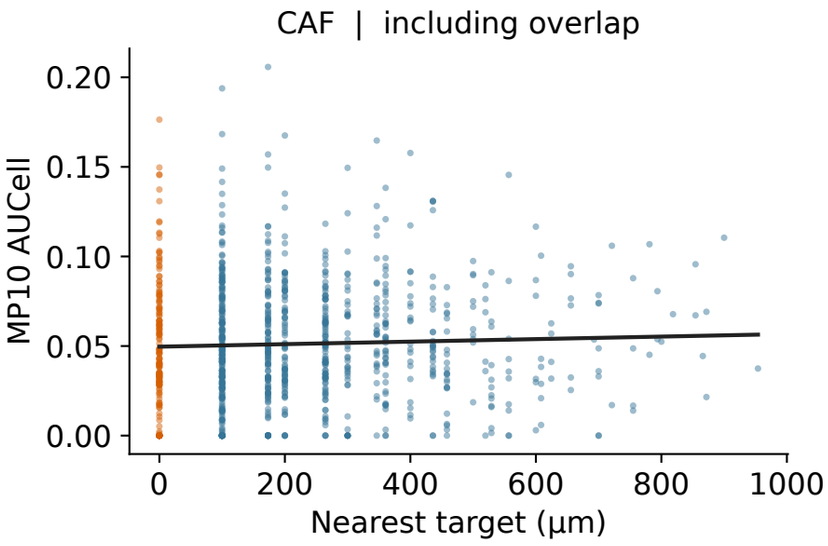
r -0.14 ρ -0.11
n 3077/3821 queries • 798 targets



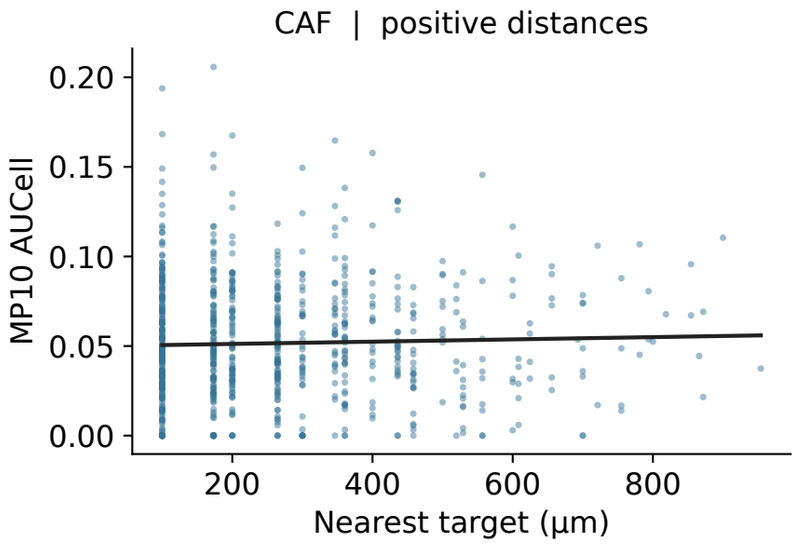
Orange = zero distance
Line = within-section linear fit

GSE171351_B1 | MP10 versus nearest target

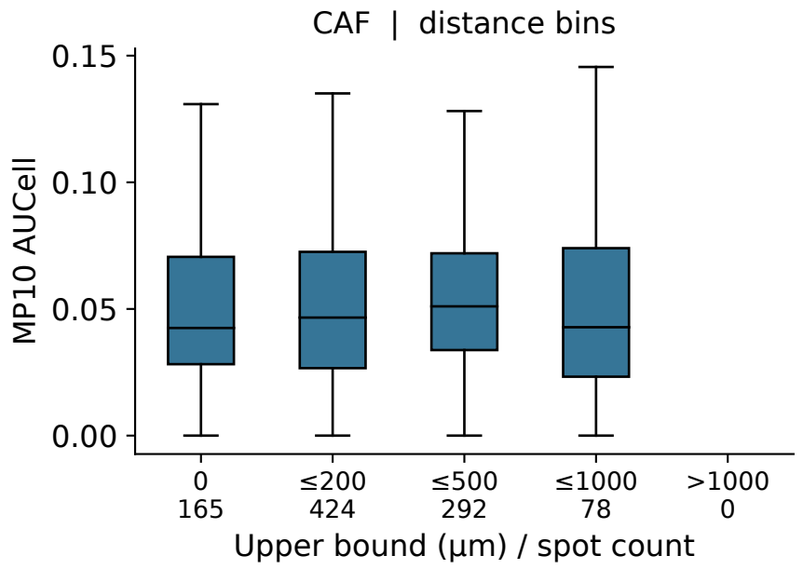
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



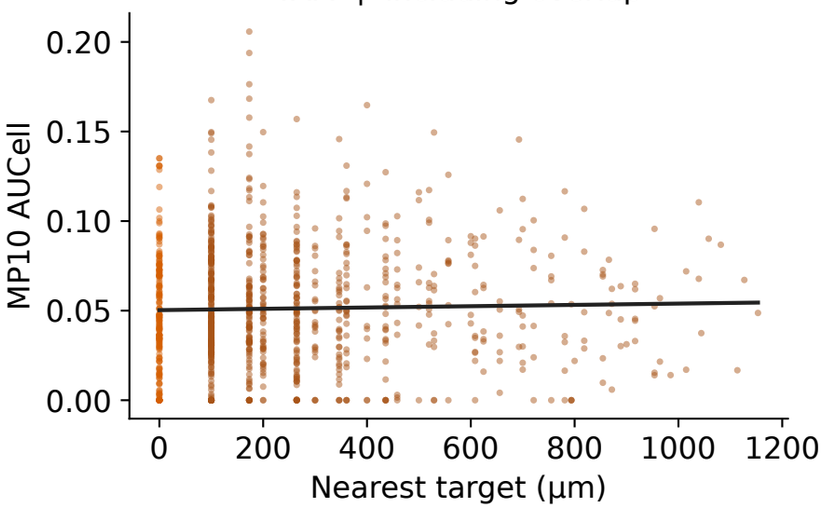
r +0.04 ρ +0.07
n 959/959 queries • 204 targets



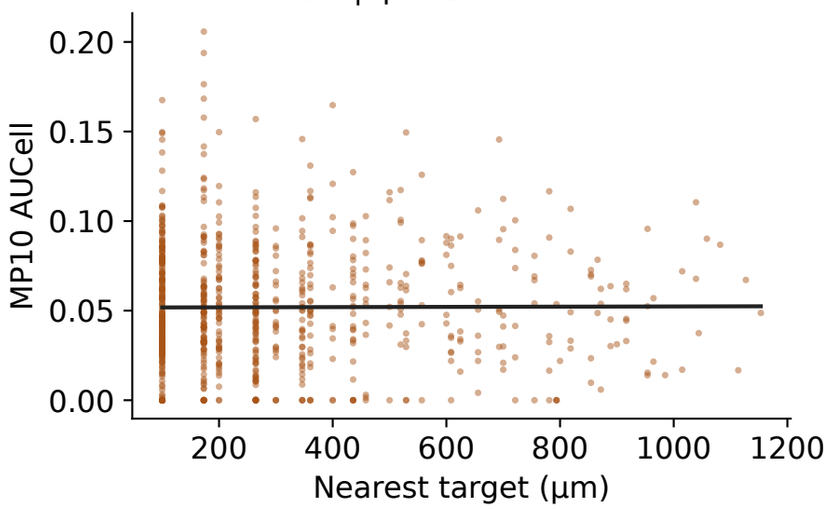
r +0.03 ρ +0.07
n 794/959 queries • 204 targets



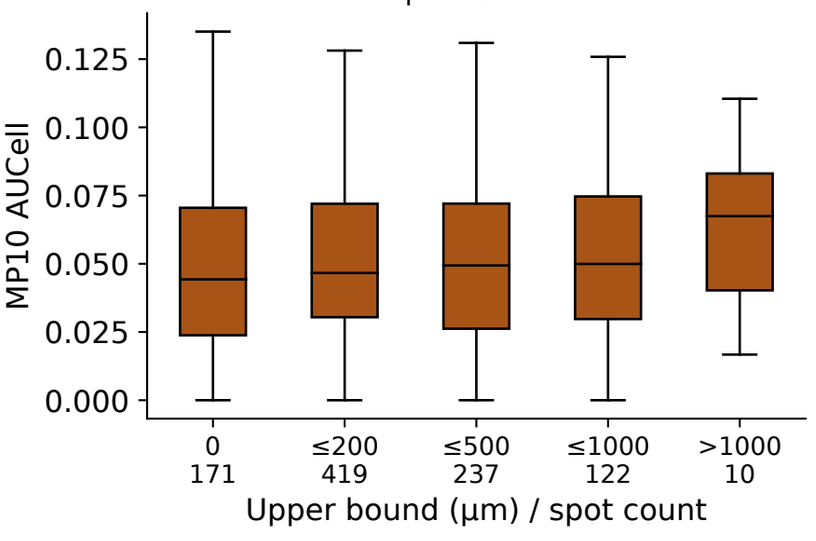
Orange = zero distance
Line = within-section linear fit



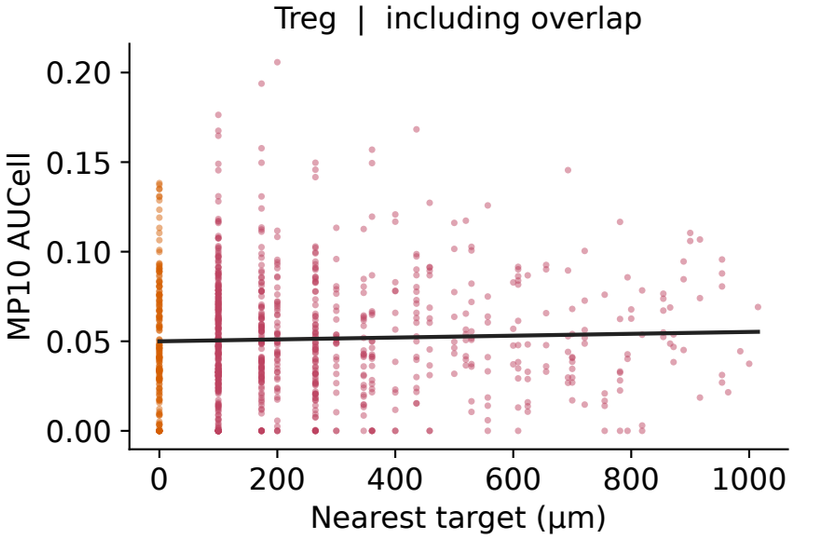
r +0.03 ρ +0.05
n 959/959 queries • 204 targets



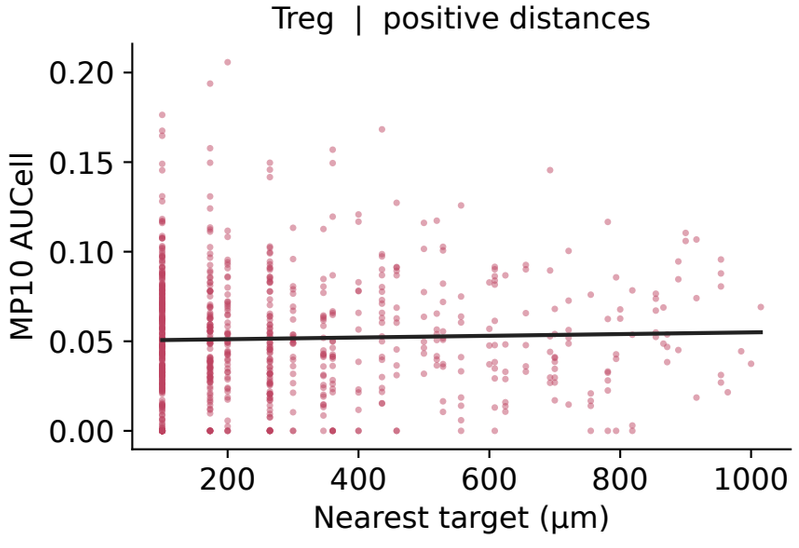
r +0.00 ρ +0.03
n 788/959 queries • 204 targets



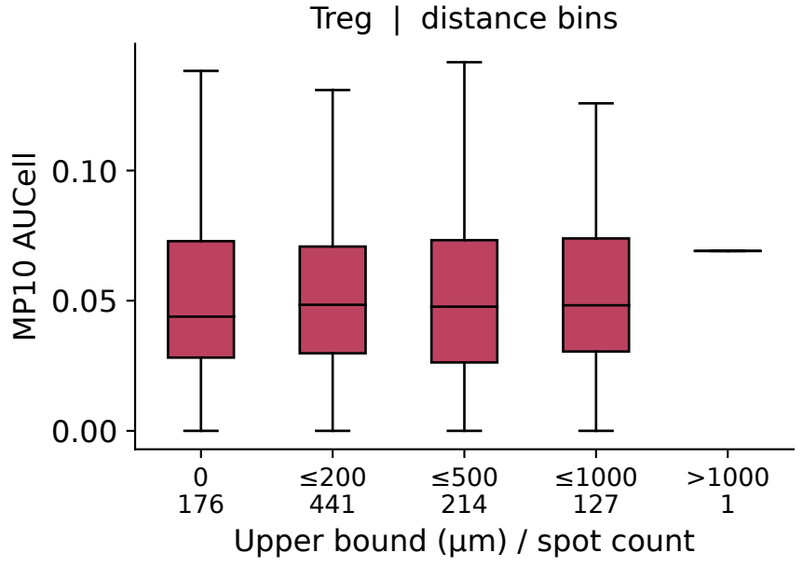
Orange = zero distance
Line = within-section linear fit



r +0.04 ρ +0.03
n 959/959 queries • 204 targets



r +0.03 ρ +0.03
n 783/959 queries • 204 targets



Orange = zero distance
Line = within-section linear fit